Memory Notes

Simplified LPDDR4 Simplified edition - JEDEC JESD209-4E

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Table of Contents

1 Simplified Lpddr4 Index....11

2 Introduction....12

2.1 What LPDDR4 SDRAM is....12

2.2 Family positioning....13

2.3 Key features of LPDDR4....14

2.4 Supplies and voltage ranges....16

2.5 Speed grades....17

2.6 The simplified drill model....18

3 Organization....19

3.1 Banks, rows, columns....19

3.2 Prefetch and IO width....20

3.3 Burst structure....21

3.4 Channels....22

3.5 The simplified drill model....23

4 Acronyms and Conventions....24

4.1 Signal names....25

4.2 Commands....26

4.3 Architecture terms....27

4.4 Training and calibration terms....28

4.5 Power and refresh terms....29

4.6 Conventions....30

5 Power-Up and Initialization....31

5.1 Cold-start sequence....32
5.2 Reset with stable power....35
5.3 Power-off sequence....36
5.4 Notes that bite....37
6 Mode Registers....38
6.1 Register map....39
6.2 Programming rules....41
6.3 MR0 - Device information....42
6.4 MR1 - Burst and write recovery....43
6.5 MR2 - Read and write latency....44
6.6 MR3 / MR11 - I/O and ODT....45
6.7 MR4 / MR13 / MR24 - Refresh, temperature, FSP and RFM....46
6.8 MR12 / MR14 - Vref settings....47
6.9 MR16 / MR17 / MR22 / MR25....48
6.10 MR32 / MR40 - DQ calibration patterns....49
7 Command Truth Tables....50
7.1 Pin conventions....51
7.2 Command truth table....52
7.3 CKE and power-state behavior....54
7.4 What is not present....55
8 Commands: Rules and Constraints....56
8.1 ACT - Activate....57
8.2 RD - Burst read....58
8.3 WR - Burst write....59

8.4 Masked Write..... 60
8.5 PRE - Precharge..... 61
8.6 Auto-Precharge..... 62
8.7 REFab / REFpb..... 63
8.8 RFM - Refresh Management..... 64
8.9 Self Refresh..... 65
8.10 MRR / MRW / MPC..... 66
8.11 Power-down..... 67
8.12 Clock stop and frequency change..... 68
8.13 NOP / Deselect..... 69
9 Walkthrough: AXI Read, Row Already Open..... 70
9.1 Scenario..... 71
9.2 Command trace..... 72
9.3 Why each command..... 73
9.4 AXI view..... 74
9.5 What to notice..... 75
10 Walkthrough: AXI Read, Wrong Row Open..... 76
10.1 Scenario..... 77
10.2 Command trace..... 78
10.3 Why each command..... 79
10.4 AXI view..... 80
10.5 What to notice..... 81
11 Walkthrough: AXI Write to an Idle Bank..... 82
11.1 Scenario..... 83

11.2 Command trace....84
11.3 Why each command....85
11.4 AXI view....86
11.5 What to notice....87
12 Walkthrough: Mixed Reads and Writes....88
12.1 Scenario....89
12.2 Command trace....90
12.3 Why each gap....92
12.4 AXI view....93
12.5 What to notice....94
13 Data Path....95
13.1 Prefetch and burst....95
13.2 Read and write strobes....96
13.3 Read and write latencies....97
13.4 WDQS control....98
13.5 Masked write, DM, and DBI....99
13.6 What LPDDR4 does not have....100
14 On-Die Termination....101
14.1 CA-bus ODT....102
14.2 DQ/DQS/DMI ODT....103
14.3 ODT during special operations....104
14.4 ZQ calibration....105
14.5 What LPDDR4 does not have....106
15 Core Timing Parameters....107

15.1 Row-cycle parameters....108
15.2 Activate-rate parameters....109
15.3 Data-to-precharge parameters....110
15.4 Why these numbers dominate....111
16 Turnaround and Burst Timing....112
16.1 Latencies and CAS-to-CAS....113
16.2 Direction changes....114
16.3 Strobe-level notes....115
17 Refresh and Self-Refresh Timing....116
17.1 Refresh parameters....116
17.2 Self-refresh....118
17.3 Power-down....119
18 Package and Electrical Notes....120
18.1 Packages and channels....121
18.2 Supplies and signaling....122
18.3 Speed grades....123
18.4 Characterization....124
19 Test Features....125
19.1 VRCG high-current VREF generator....126
19.2 CA and DQ VREF training....127
19.3 Command bus training....128
19.4 Frequency Set Points....129
19.5 Write leveling....130
19.6 RD DQ calibration and read preamble training....131

19.7 DQS-DQ training....132
19.8 DQS interval oscillator....133
19.9 Multi-purpose command....134
19.10 Thermal offset and temperature sensor....135
19.11 ZQ calibration....136
19.12 Refresh management....137
19.13 Post-package repair....138
19.14 What is not here....139

List of Figures

No figures in this document.

List of Tables

No tables in this document.

List of Waveforms

No waveforms in this document.

1 Simplified Lpddr4 Index

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2 Introduction

2.1 What LPDDR4 SDRAM is

LPDDR4 SDRAM is the fourth generation of JEDEC low-power double-data-rate memory, standardized as JESD209-4E. It is built for mobile and battery-powered systems, usually in package-on-package (PoP) or discrete FBGA packages. The interface operates at roughly 1.1 V, lower than the 1.2 V of LPDDR3, and adds features aimed at higher bandwidth, lower power, and easier DVFS transitions.

The command/address path uses a 6-bit CA bus. Commands, row/column addresses and bank select information are transferred on the rising clock edge across CA0-CA5 over one, two or four clock cycles. CS_n and CKE remain single-edge control inputs. A dedicated RESET_n pin is provided, unlike LPDDR3 where reset was only available through an MRW command.

2.2 Family positioning

Tech	Prefetch	IO voltage	DLL	Command/ address style
LPDDR2 (JESD209-2)	4n (S4) / 2n (S2)	1.2 V	No	10-bit DDR CA bus
LPDDR3 (JESD209-3)	8n	1.2 V	No	10-bit DDR CA bus
LPDDR4 (JESD209-4)	16n	1.1 V	No	6-bit DDR CA bus

LPDDR4 is therefore the first low-power DDR generation to move to a 16n prefetch, BL16 as the primary burst, and a dual-channel-per-die organization.

2.3 Key features of LPDDR4

- Dual channel per die: a standard LPDDR4 die contains two fully independent channels, each with its own CA bus, clock, CKE, CS_n and DQ/DQS/DMI pins. This book treats one channel at a time; channel B behaves identically and independently.
- 16n prefetch: a column access moves sixteen words internally per DQ. With two words transferred per DQ per clock, the minimum burst is eight clocks. This is why tCCD = 8 clocks for BL16.
- Burst length: BL16 is the baseline. BL32 and on-the-fly BL16/BL32 are also programmable through MR1.
- 8 banks flat per channel, addressed by BA0-BA2. There are no bank groups and no stack IDs.
- x16 and x8 byte-mode data widths per channel; byte-mode dies can be combined into a x16 configuration.
- No DLL. Read data timing is specified as a tDQSCK window relative to the clock.
- Read latency (RL) and write latency (WL) are programmed through mode registers (MR1/MR2), not via additive latency.
- Dedicated RESET_n pin for hardware reset.
- Masked Write command (MWR) allows byte-lane masking on writes without a separate DM pin per byte. The DMI pin carries mask information during masked writes.
- DMI pins double as data mask (DM) and data bus inversion (DBIdc) signals.
- WDQS control modes let the controller manage DQS stability around writes and masked writes.
- Frequency Set Points (FSP) store two complete timing/register profiles and allow fast DVFS switching.
- Refresh Management (RFM) command mitigates row-address-access disturbance.
- Multi-Purpose Command (MPC) provides training, FIFO, oscillator and ZQ calibration operations through a single command opcode.
- CA-bus ODT plus DQ/DQS/DMI ODT for command/address and data termination.
- Vref training for both CA (MR12) and DQ (MR14).
- Command bus training to align CA/CS with CK and set internal VREF(CA).

- ZQ calibration and ZQCal reset for output driver and termination calibration.
- Post-package repair (PPR) to replace a defective row with a spare row after assembly.
- DQS interval oscillator for tracking clock-tree delay drift over temperature and voltage.
- Per-bank refresh (REFpb) and all-bank refresh (REFab) both exist.

2.4 Supplies and voltage ranges

Supply	Nominal	Function
VDD1	1.80 V	Core power
VDD2	1.10 V	Core and input-buffer power
VDDQ	1.10 V	DQ/DQS/DMI I/O power

Vref inputs are generated and trained internally rather than supplied as external reference pins.

2.5 Speed grades

Data-rate grades cover 533 through 4267 MT/s, with clock frequencies from above 10 MHz up to 2133 MHz. RL and WL are selected from programmed tables that map counts to frequency ranges. Core timings such as tRCD, tRP and tRAS are given in nanoseconds, then converted to clock cycles by the controller.

2.6 The simplified drill model

Real parts have thousands of rows and hundreds of columns. All examples in this book use a deliberately tiny geometry so bank state is obvious at a glance:

- Banks: B0-B7 (eight, as on every LPDDR4 channel)
- Rows: R0-R7 per bank
- Columns: C0-C7 per row

Because the model only has eight columns, a BL16 access wraps the column space twice: the burst starts at the issued column, visits C0-C7, then wraps back to C0 and continues. Real parts use 64 fetch boundaries and many more columns; the scheduling rules are identical.

Source: JESD209-4E sections 1, 2.1, 3.1, 4.12, 6.1

3 Organization

3.1 Banks, rows, columns

An LPDDR4 channel is organized as eight independent banks of DRAM arrays. Each bank has its own sense-amplifier row buffer, so up to eight rows can be open at the same time within one channel. The bank count is always eight; the spec does not define 4-bank LPDDR4 SDRAM devices.

Address geometry per density (from the spec addressing tables):

Density	Channels	Banks	Rows (x16)	Columns	Page size (x16)
2 Gb	2	8	R0-R12	C0-C9	2048 B
4 Gb	2	8	R0-R13	C0-C9	2048 B
6 Gb	2	8	R0-R14*	C0-C9	2048 B
8 Gb	2	8	R0-R14	C0-C9	2048 B
12 Gb	2	8	R0-R15*	C0-C9	2048 B
16 Gb	2	8	R0-R15	C0-C9	2048 B
24 Gb	2	8	R0-R16*	C0-C9	2048 B
32 Gb	2	8	R0-R16	C0-C9	2048 B

*For non-binary densities, the top quarter of the row address space is invalid: when the most-significant row address bit is high, the next bit must be low.

Single-channel dies carry the same per-channel geometry but only one channel; dual-channel dies contain two independent copies of the table above. Byte-mode (x8) parts halve the prefetch and page size while doubling the row count for the same die density.

Addressing rules:

- The two least-significant column bits (C0-C1) are implied zero and are not transmitted on the CA bus. Column addresses on the bus start at C2.
- Unused row/column address bits on the CA bus must still be driven to valid logic levels.
- REFpb can only target a bank that is currently idle and precharged.

3.2 Prefetch and IO width

- 16n prefetch: a column access fetches sixteen words per DQ internally. The DDR interface delivers two words per DQ per clock, so a BL16 burst occupies eight clocks. This is why tCCD = 8 clocks for BL16 and 16 clocks for BL32.
- Data bus widths are x16 and x8 (byte mode). Each byte lane has its own differential strobe (DQS_t/DQS_c) and its own DMI pin: one DQS pair and one DMI per 8 DQs.
- DMI replaces the older DM pin. It carries data-mask information for masked writes and data-bus-inversion status for reads and normal writes when DBIdc is enabled.

3.3 Burst structure

Accesses are burst-oriented: each Read, Write or Masked Write command transfers BL words per DQ, starting at the issued column and following the programmed burst order. BL is normally 16; BL32 and on-the-fly BL16/BL32 are also available. Burst addressing follows the standard sequential pattern driven by the low-order column bits.

3.4 Channels

A standard LPDDR4 die contains two fully independent channels (channel A and channel B). Each channel has its own CA bus, clock, CKE, CS_n, ODT(ca), DQ, DQS and DMI pins. The two channels share only package-level pins such as RESET_n and ZQ. The protocol in this book is per-channel; the other channel operates independently with the same rules.

3.5 The simplified drill model

Per the Introduction, all examples use eight banks (B0-B7), eight rows (R0-R7) and eight columns (C0-C7). Real parts are far larger; the small model exists to make bank/row state obvious in command traces. A BL16 access wraps through the eight-column model twice.

Source: JESD209-4E sections 2.1, 3.1

4 Acronyms and Conventions

Terms used precisely in this book.

4.1 Signal names

Term	Meaning
CK_t / CK_c	Differential clock. Positive edge = CK_t rising through CK_c falling
CKE	Clock enable. SDR, sampled on the positive clock edge; gates all power states
CS_n	Chip select. SDR, sampled on the positive clock edge; contributes to the command code
CA0-CA5	6-bit command/address bus; sampled on the rising clock edge
DQ	Bidirectional data
DQS_t / DQS_c	Differential data strobe, one pair per byte lane; edge-aligned on reads, center-aligned on writes
DMI	Data mask / data bus inversion pin, one per byte lane
ODT(ca)	On-die termination control for the CA bus
ODT	On-die termination for DQ/DQS/DMI
RESET_n	Active-low hardware reset pin
ZQ	Pin used as reference for output drive-strength and ODT calibration

4.2 Commands

Term	Meaning
ACT	Activate (open a row in a bank)
RD / WR	Burst read / burst write
MWR	Masked write (write with byte-lane masking)
PRE	Precharge (close a row); per-bank or all-bank via the AB flag
REFab / REFpb	Refresh, all banks / one per-bank-refresh bank
RFM	Refresh management command
MRR / MRW	Mode register read / mode register write
MPC	Multi-purpose command (training, FIFO, ZQ, oscillator)
NOP	No operation
SREF / PD / DPD	Self-refresh, power-down, deep power-down

4.3 Architecture terms

Term	Meaning
Prefetch (16n)	Words moved internally per column access; LPDDR4 = 16n
BL	Burst length; BL16 default, BL32 and on-the-fly optional
RL / WL	Read latency / write latency, in clocks, programmed in MR1/MR2
nWR	Write recovery in clocks for auto-precharge, programmed in MR1
nRTP	Read-to-precharge delay in clocks for auto-precharge
AP	Auto-precharge flag
AB	All-bank flag
RU{}	Round-up function; RU{x} = smallest integer >= x
RD{}	Round-down function; RD{x} = largest integer <= x
FSP	Frequency set point; two complete register sets for fast DVFS switching
FSP-OP	Selects which frequency set point is currently active
FSP-WR	Selects which frequency set point is accessed by MRW/MRR
DBIdc	Data bus inversion for power saving and signal integrity
WDQS	DQS control mode for write and masked write stability
PPR	Post-package repair of a failed row

4.4 Training and calibration terms

Term	Meaning
CA training / CBT	Command bus training; aligns CA/CS with CK and sets VREF(CA)
VREF(CA)	Internal reference voltage for CA inputs, trained via MR12
VREF(DQ)	Internal reference voltage for DQ inputs, trained via MR14
VRCG	VREF current generator, enabled for faster VREF settling
Write leveling	Training mode to align DQS with the clock; entered/exited via MRW
DQ calibration	MPC reads return predefined calibration patterns from FIFOs
ZQCal	Output driver and termination calibration using the ZQ pin

4.5 Power and refresh terms

Term	Meaning
PASR	Partial array self-refresh: refresh only masked banks/segments in self-refresh
Refresh Rate	TCSR refresh multiplier from MR4
DAI	Device auto-initialization status bit (MR0 OP0)
TUF	Temperature update flag (MR4 OP7)
DPD	Deep power-down; array contents lost, full re-init required on exit
tREFW / tREFI / tREFIM	Refresh window / average refresh interval / multiplied refresh interval
tRFCab / tRFCpb	Refresh cycle time, all-bank / per-bank
RZQ	240 ohm reference resistor for output driver and ODT calibration

4.6 Conventions

- H/L in truth tables are logic high/low at the sampled edge; X is “don’t care” but must still be a valid logic level.
- Timing symbols are written as plain text: tRCD, tWTR, tREFIpb.
- tRTW is the book symbol for the read-to-write turnaround rule. The spec names the rule tRTW in the MPC timing table; the same symbol is used for the general RD->WR spacing in later timing tables.
- Anything the spec leaves unspecified is illegal; after an illegal event the part has to be reset through a full power-down/re-init cycle.

Source: JESD209-4E sections 2.4 (Tables 1-2), 3.4, 4.12, 4.16, 4.26-4.28, 4.35, 4.38, 4.47-4.48

5 Power-Up and Initialization

LPDDR4 must follow a fixed bring-up sequence; any deviation is undefined. Each channel initializes independently, though a single RESET_n pin resets every channel on the die. This chapter describes one channel; repeat the same steps for the other.

5.1 Cold-start sequence

1. **Voltage ramp (Ta -> Tb).** Apply VDD1, VDD2 and VDDQ while RESET_n is held LOW (<= 0.2 x VDD2) and CKE is LOW. Ta is the first moment any supply reaches 300 mV; Tb is when every supply and reference is inside its operating range. The ramp must finish within tINIT0 = 20 ms. During the ramp: VDD1 must stay at or above VDD2; VDD2 must stay at or above VDDQ - 200 mV; VSS/VSSQ/VSSCA must stay within 100 mV of each other; and DQ/DMI/DQS_t/DQS_c must stay between VSSQ and VDDQ while CKE/CS/CA/CK must stay between VSS and VDD2.
2. **RESET_n hold (Tb -> Tc).** Keep RESET_n LOW for at least tINIT1 = 200 us after Tb. At least 10 ns before releasing RESET_n, drive CKE LOW.
3. **Clock and CKE (Tc -> Td).** After RESET_n rises, wait at least tINIT3 = 2 ms before taking CKE HIGH. The differential clock must be stable for at least tINIT4 = 5 tCK before CKE goes HIGH. CS must be LOW when CKE rises.
4. **Idle wait (Td -> Te).** With CKE HIGH, issue only DES (NOP) for at least tINIT5 = 2 us. The clock must remain inside the boot window tCKb during MRW and MRR commands issued here; some AC parameters such as tDQSCK are relaxed until full configuration.
5. **Configure I/O (Te -> Tf).** Issue MRW commands to set pull-up calibration point (MR3 PU-CAL), pull-down drive strength (MR3 PDDS), DQ ODT (MR11 OP[2:0]) and CA ODT (MR11 OP[6:4]). These values are needed before ZQ calibration latches.
6. **ZQ calibration (Tf -> Th).** Issue the MPC ZQCal Start command. Wait tZQCAL = 1 us, then issue the MPC ZQCal Latch command. Keep the CA bus deselected for tZQLAT = max(30 ns, 8 tCK) so the latched values update the DQ drivers and CA/DQ termination. Do not overlap ZQCal Start commands across devices that share one ZQ resistor.
7. **Command bus training (Th).** Enter command-bus training (MR13 OP[0] = 1) to align CS/CA with CK and set the internal VREF(CA) level. The device powers up with low-speed receiver defaults, so speeds above tCKb require this step.
8. **Write leveling (Ti).** Enable write leveling (MR2 OP[7] = 1) and adjust DQS_t/DQS_c timing so the device recognizes the start of a write burst at the programmed WL. Clear MR2 OP[7] to exit.

9. **DQ bus training (Tj).** Use MPC Read FIFO, Write FIFO and Read DQ Calibration together with MRW updates to MR14 to train VREF(DQ), DQS and DQ timing.
10. **Normal operation (Tk).** The channel is now ready for any valid command. Write any remaining operating mode registers (burst length, RL/WL, DBI, etc.).

```
ramp <=20ms   RESET 200us   RESET_n ->    CKE ->      DES 2us     MRW
IO cfg    MPC ZQStart   MPC ZQLatch    CBT      WL      DQ train
ready
|___tINIT0____|__tINIT1_____|__tINIT2_____|__tINIT3___|__tINIT5___|
____Te___|____tZQCAL____|___tZQLAT_____|__Th___|__Ti___|____Tj____|
__Tk__
                          tINIT2=10ns CKE low before RESET_n high
                          tINIT4=5tCK stable clock before CKE high
```

Parameter	Min	Max	Unit	Meaning
tINIT0	-	20	ms	longest allowed supply ramp
tINIT1	200	-	us	RESET_n LOW after all supplies valid
tINIT2	10	-	ns	CKE LOW before RESET_n rises
tINIT3	2	-	ms	CKE LOW after RESET_n rises
tINIT4	5	-	tCK	stable clock before CKE rises
tINIT5	2	-	us	DES wait after CKE goes HIGH

Parameter	Min	Max	Unit	Meaning
tZQCAL	1	-	us	ZQ calibration time
tZQLAT	max(30ns, 8tCK)	-	ns	quiet CA bus after ZQCal Latch
tCKb	18	note	ns	boot clock period; system may start faster

5.2 Reset with stable power

Drive RESET_n below 0.2 x VDD2 for at least tPW_RESET = 100 ns, with CKE LOW at least 10 ns before RESET_n rises. Then repeat steps 3-10 of the cold-start sequence. Mode registers return to defaults and SDRAM contents become undefined.

5.3 Power-off sequence

Controlled power-off reverses the ramp rules. Hold CKE LOW (<= 0.2 x VDD2) and keep all other inputs between VILmin and VIHmax; outputs remain High-Z. Tx is when any supply first drops below its minimum; Tz is when all supplies are below 300 mV. Between Tx and Tz, VDD1 must stay above VDD2 and VDD2 must stay above VDDQ - 200 mV; VSS pins must stay within 100 mV. The fall from Tx to Tz must not exceed tPOFF = 2 s.

Uncontrolled power-off is tolerated only with tighter limits: at Tx all supplies must turn off and supply current capacity must reach zero except for residual charge; VDD1 and VDD2 must fall with slope <= 0.5 V/us between Tx and Tz; and this may occur at most 400 times over the device's life.

5.4 Notes that bite

- LPDDR4 has no DAI bit; the device is ready for MRW/MRR after tINIT5. Do not poll MR0 for an initialization-done flag.
- Boot-frequency AC relaxations apply until command-bus, write-leveling and DQ-bus training are complete.
- ZQCal Start may be issued to either or both channels of a dual-channel die, but ZQCal Latch is required per channel.
- Deep power-down exit requires the full power-up initialization sequence.

Source: JESD209-4E sections 3.3.1, 3.3.2, 3.3.3, 3.3.4

6 Mode Registers

LPDDR4 uses MRW (write) and MRR (read) commands. Both carry an 8-bit register address (MA0-MA7) and an 8-bit operand (OP0-OP7). MRW requires all banks idle; MRR is allowed from all-banks-idle or banks-active states. RFU bits are written 0 and read as 0.

6.1 Register map

MR	Name	Access	Highlights
MR0	Device Info	R	Refresh mode, latency mode, RFM support, ZQ pin self-test, CA terminating rank, scaling support
MR1	Device Feature 1	W	BL, nWR, read/write preamble, post-amble
MR2	RL / WL	W	RL, WL, WL set select, write-leveling enable
MR3	I/O Config	W	DBI-WR/RD, PDDS, PPR protect, write post-amble, PU-CAL
MR4	Refresh / Temp	R/W	Refresh rate, TUF, SR abort, PPR entry, thermal offset
MR5-MR7	IDs	R	Manufacturer and revision IDs
MR8	Basic Config	R	Type (S16, 16n prefetch), density, IO width
MR10	ZQ Reset	W	OP[0]=1 resets calibration to defaults
MR11	ODT Control	W	DQ ODT (OP[2:0]), CA ODT (OP[6:4])
MR12	VREF(CA)	R/W	Range and value for CA Vref

MR	Name	Access	Highlights
MR13	FSP / Training	W	FSP-OP/WR, DMD, RRO, VRCG, VRO, RPT, CBT
MR14	VREF(DQ)	R/W	Range and value for DQ Vref
MR16	PASR Bank Mask	W	Per-bank self-refresh mask
MR17	PASR Segment Mask	W	Per-row-segment self-refresh mask
MR18-MR19	DQS Oscillator	R	Counter result from MPC Start DQS Osc
MR22	ODT Details	W	SoC ODT, ODTE-CK/CS, ODTD-CA, byte-mode disable
MR23	DQS Timer	W	Auto-stop interval for DQS oscillator
MR24	RFM	R	RFM required, RAAIMT, RAAMMT
MR25	PPR Resource	R	Per-bank PPR resource availability
MR26	Scaling Level	R	Vendor scaling level (valid if MR0 OP[6]=1)
MR32 / MR40	DQ Cal Patterns	W	Patterns returned by MPC Read DQ Calibration

Many parameters above exist in two physical copies, one per Frequency Set Point. MR13 OP[6] (FSP-WR) selects which copy is accessed by MRW/MRR; MR13 OP[7] (FSP-OP) selects which copy drives current operation.

6.2 Programming rules

Symbol	Definition	Min
tMRR	Gap occupied by an MRR (only DES allowed inside)	8 tCK
tMRW	Gap occupied by an MRW (only DES allowed inside)	max(10 ns, 10 tCK)
tMRD	Wait from MRW registration to next non-MRW command	max(14 ns, 10 tCK)
tMRRI	Extra wait after power-down exit before MRR	tRCD + 3 tCK
tMPCWR	Extra wait after power-down exit before MPC Write FIFO	tRCD + 3 tCK

MPC read/write training commands are followed immediately by a CAS-2 command whose operands are driven LOW. MPC ZQCal Start/Latch, DQS oscillator start/stop and NOP do not need CAS-2.

6.3 MR0 - Device information

Field	Bits	Meaning
Refresh mode	OP[0]	0 = legacy and modified refresh; 1 = modified refresh only
Latency mode	OP[1]	0 = x16 latency; 1 = byte-mode latency
RFM support	OP[2]	0 = TRR supported; 1 = RFM supported
RZQI	OP[4:3]	ZQ-pin self-test result (read after ZQ calibration)
SCLS	OP[6]	1 = scaling level supported
CATR	OP[7]	1 = this rank terminates CA (vendor specific)

6.4 MR1 - Burst and write recovery

Field	Bits	Encodings
BL	OP[1:0]	00 = BL16; 01 = BL32; 10 = BL16/32 on-the-fly
WR-PRE	OP[2]	0 = reserved; 1 = 2 tCK write preamble
RD-PRE	OP[3]	0 = static; 1 = toggling read preamble
nWR	OP[6:4]	Programmed write-recovery for auto-precharge (encoding depends on MR26 scaling)
RPST	OP[7]	0 = 0.5 tCK read post-amble; 1 = 1.5 tCK

6.5 MR2 - Read and write latency

OP[2:0]	RL (x16, DBI-RD off)	WL (Set A / Set B)
000	6 (default)	4 / 4
001	10	6 / 8
010	14	8 / 12
011	20	10 / 18
100	24	12 / 22
101	28	14 / 26
110	32	16 / 30
111	36	18 / 34

OP[6] selects WL Set A (0) or B (1). OP[7] enables write leveling. DBI-RD on in MR3 adds latency to RL (see spec tables).

6.6 MR3 / MR11 - I/O and ODT

MR3: PDDS sets pull-down drive strength and Rx termination (RZQ/1 through RZQ/6). DBI-WR and DBI-RD enable data-bus inversion. PU-CAL selects VDDQ/2.5 or VDDQ/3 output-high calibration point.

MR11: DQ ODT (OP[2:0]) and CA ODT (OP[6:4]) each choose disable or RZQ/n.

6.7 MR4 / MR13 / MR24 - Refresh, temperature, FSP and RFM

MR4 OP[2:0] reports the temperature-compensated refresh multiplier (4x down to 0.25x with/without de-rating). OP[7] (TUF) is set when the refresh code has changed since the last MRR to MR4. OP[4] enters PPR mode; OP[3] enables self-refresh abort.

MR13 OP[7:6] are FSP-OP and FSP-WR. OP[5] disables data masking when set. OP[4] enables additional refresh-rate codes. OP[3] speeds VREF settling; OP[0] enters command-bus training.

MR24 reports whether RFM is required (OP[0]), the initial RAA threshold (OP[5:1]) and the maximum RAA multiplier (OP[7:6]).

6.8 MR12 / MR14 - Vref settings

MR12 controls VREF(CA); MR14 controls VREF(DQ). Each has a range bit (OP[6]) and a 6-bit value (OP[5:0]). Range[0] spans roughly 10%-30% of VDD2/VDDQ; Range[1] spans roughly 22%-42%.

6.9 MR16 / MR17 / MR22 / MR25

- MR16 OP[7:0] masks individual banks from self-refresh.
- MR17 OP[7:0] masks row segments from self-refresh.
- MR22 fine-tunes command-bus termination: SoC ODT value, CK/CS ODT override for non-terminating ranks, and per-byte CA/CK/CS ODT disable in byte mode.
- MR25 OP[7:0] reads back per-bank PPR resource availability.

6.10 MR32 / MR40 - DQ calibration patterns

MPC Read DQ Calibration returns MR32 (default 0x5A) followed by MR40 (default 0x3C) on the DQ bus. MR15 and MR20 invert the pattern on selected lower/upper byte DQ pins.

Source: JESD209-4E sections 3.4.1, 4.23, 4.24, 4.29, 4.35, 4.38, 4.47, 4.48

7 Command Truth Tables

LPDDR4 packs two independent channels per die; each channel has eight banks and no bank groups. There is no DLL, so read and write latencies are programmed through mode registers. Commands and addresses travel over the 6-bit CA bus across two clock cycles; a RESET_n pin is provided for full device reset. Examples in this book use the drill model (banks B0-B7, rows R0-R7, columns C0-C7) on one channel; the other channel is independent.

7.1 Pin conventions

- CKE is sampled at rising CK edges and must be high for any command in the table below.
- CS is active high: a two-cycle command starts with CS high on the first rising edge and CS low on the second; Deselect is one cycle with CS low.
- CAxr is CA bit x on the first rising edge; CAxf is the same bit on the second rising edge.
- H/L are valid logic levels; V means a defined level whose value depends on the command; X means the pin may be floated.

7.2 Command truth table

Command	First cycle CA0-CA5	Second cycle CA0-CA5	Notes
Deselect (DES)	X	(none)	One cycle, CS = L
Multi-Purpose (MPC)	L L L L L OP6	OP0 OP1 OP2 OP3 OP4 OP5	NOP when OP6 = 0; training when OP6 = 1
Precharge (PREpb/PREab)	L L L L H AB	BA0 BA1 BA2 V V V	AB = 1 closes all banks; AB = 0 closes the selected bank
Refresh (REFpb/REFab)	L L L H L AB	BA0 BA1 BA2 RFM V V	RFM bit is valid only when MR24 OP[0] = 1
Self Refresh Entry (SRE)	L L L H H V	V V V V V V	CKE falls during SRE to enter
Write-1 (WR-1)	L L H L L BL	BA0 BA1 BA2 V C9 AP	Must be followed immediately by CAS-2
Self Refresh Exit (SRX)	L L H L H V	V V V V V V	CKE rises to exit
Mask Write-1 (MWR-1)	L L H H L L	BA0 BA1 BA2 V C9 AP	BL is fixed at 16; followed by CAS-2
Read-1 (RD-1)	L H L L L BL	BA0 BA1 BA2 V C9 AP	Must be followed immediately by CAS-2
CAS-2 (WR-2/MWR-2/RD-2/MRR-2/MPC data)	L H L L H C8	C2 C3 C4 C5 C6 C7	Carries remaining column bits; C1-C0 are implied zero
Mode Register Write-1 (MRW-1)	L H H L L OP7	MA0 MA1 MA2 MA3 MA4 MA5	Must be followed immediately by MRW-2
Mode Register Write-2 (MRW-2)	L H H L H OP6	OP0 OP1 OP2 OP3 OP4 OP5	Second half of MRW data

Command	First cycle CA0-CA5	Second cycle CA0-CA5	Notes
Mode Register Read-1 (MRR-1)	L H H H L V	MA0 MA1 MA2 MA3 MA4 MA5	Must be followed immediately by CAS-2
Activate-1 (ACT-1)	H L, then row and bank bits	BA0-BA2 plus row bits	Must be followed immediately by ACT-2
Activate-2 (ACT-2)	H H, then remaining row bits	Remaining row bits	Completes row activation

Load-bearing notes:

- Every command except DES spans two cycles and is defined by CA[5:0] sampled at the first rising edge while CS is high.
- WR-1, MWR-1, RD-1, MRR-1, and MPC read/write training commands must be immediately followed by CAS-2 with no command in between.
- ACT-1 must be immediately followed by ACT-2; MRW-1 must be immediately followed by MRW-2.
- The BL bit in WR-1/RD-1 selects burst length on-the-fly when that feature is enabled: low = BL16, high = BL32. LPDDR4 supports BL16 and BL32; the native prefetch is 16n.
- Auto-precharge is selected by AP = 1 in WR-1/MWR-1/RD-1.
- MWR-1 supports only BL16; in a BL32 configuration the controller supplies only a 16-bit data window for the masked write.

7.3 CKE and power-state behavior

Condition	Result
CKE falls while CS is low (DES on bus)	Enter power-down
CKE falls with SRE command on bus	Enter self refresh
CKE rises from power-down	Exit after tXP
CKE rises from self refresh	Exit after tXSR; abort path available via MR4 OP[3]
CKE falls during RD, WR, MWR, MRR, MRW, VREF MRW, CBT MRW, VRCG MRW, or Start DQS Oscillator	Illegal

7.4 What is not present

LPDDR4 has no single-cycle NOP command (MPC with OP6 = 0 acts as NOP), no burst terminate, no burst chop, no bank groups, and no MRS command; mode-register updates use MRW.

Source: JESD209-4E sections 4.1, 4.2, 4.3, 4.6, 4.10, 4.15, 4.16, 4.17, 4.19, 4.20, 4.23, 4.24, 4.35, 4.44, 4.45, 4.46, 4.47

8 Commands: Rules and Constraints

LPDDR4 uses two-cycle commands on a 6-bit CA bus. Each die contains two independent channels; the examples here use one channel with banks B0-B7. The device has no DLL and no bank groups. Burst length is BL16 or BL32 on-the-fly; masked writes are BL16 only. Read latency (RL) and write latency (WL) are programmed through mode registers.

8.1 ACT - Activate

Opens row R in bank B. The command is issued as ACT-1 followed immediately by ACT-2; the row address is split across both cycles and the bank address is carried in ACT-1.

- RD or WR to the same bank may follow after tRCD.
- ACT to a different bank needs tRRD; ACT to the same bank again needs a PRE first and must satisfy tRAS (ACT-to-PRE) and tRC (ACT-to-ACT).
- On 8-bank devices, no more than four ACT or REFpb operations may fall inside a rolling tFAW window.

Core timing (x16 mode):

Symbol	Rule
tRCD	max(18 ns, 4 nCK)
tRRD	max(10 ns, 4 nCK), relaxed to max(7.5 ns, 4 nCK) at 4267 Mbps
tFAW	40 ns at most data rates, 30 ns at 4267 Mbps
tRAS	min = max(42 ns, 3 nCK); max = min(9 x RefreshRate x tREFI, 70.2 us)
tRC	tRAS + tRPab (with PREab) or tRAS + tRPpb (with PREpb)

8.2 RD - Burst read

Starts a BL16 or BL32 read burst from the open row. RD-1 is immediately followed by CAS-2; the starting column is a multiple of four because C1-C0 are implied zero.

- RL is programmed in the mode registers. Valid data begins RL x tCK + tDQSCK + tDQSQ after the second rising edge of the read command.
- Back-to-back reads are spaced by tCCD = 8 clocks for BL16 and 16 clocks for BL32.
- A PRE to the same bank may follow after BL/2 + max(8, RU(tRTP/tCK)) - 8 clocks, and tRAS must also be satisfied.
- The earliest WR or MWR after a RD is governed by tRTW (the spec names this symbol). With DQ ODT disabled: tRTW = RL + RU(tDQSCKmax/tCK) + BL/2 - WL + tWPRE + RD(tRPST). With DQ ODT enabled the formula includes ODT turn-on latency.

8.3 WR - Burst write

Starts a BL16 or BL32 write burst. WR-1 is immediately followed by CAS-2. The DQS strobe must arrive before DQ by tDQS2DQ and be center-aligned with the data.

- WL is programmed in the mode registers. The first latching DQS edge occurs WL x tCK + tDQSS after the command completes.
- Back-to-back writes are spaced by tCCD = 8 clocks (BL16) or 16 clocks (BL32).
- A PRE to the same bank may follow after WL + 1 + BL/2 + RU(tWR/tCK) clocks.
- The earliest RD after a WR is WL + 1 + BL/2 + RU(tWTR/tCK) clocks after the WR command.

Core timing:

Symbol	Rule (x16 mode)
tWR	max(18 ns, 6 nCK)
tWTR	max(10 ns, 8 nCK)

8.4 Masked Write

Any write that masks one or more bytes within the burst must use the Masked Write command (MWR-1 + CAS-2). Masked writes are BL16 only.

- One DMI pin per byte lane carries the mask; DMI high masks the corresponding beat across the byte, DMI low writes normally. DMI timing matches DQ timing.
- DM is enabled by MR13 OP[5]. Write DBIdc and read DBIdc are controlled separately by MR3 OP[7] and OP[6].
- Two masked writes to the same bank must be separated by tCCDMW = 32 clocks (four tCCD periods at BL16).

8.5 PRE - Precharge

Closes the open row of one bank or all banks. PRE uses AB = 1 for all-bank precharge; AB = 0 closes the bank selected by BA[2:0].

- After a per-bank PRE wait tRPpb = max(18 ns, 4 nCK); after PREab wait tRPab = max(21 ns, 4 nCK).
- PRE must not violate tRAS after ACT, tRTP after RD, or tWR after WR.

8.6 Auto-Precharge

Setting AP = 1 in RD-1, WR-1, or MWR-1 schedules an internal precharge at the earliest possible moment. The controller must still satisfy tRC before the next ACT to the same bank. RAS lockout ensures tRAS is not violated.

8.7 REFab / REFpb

Refresh all banks (REFab) or one bank (REFpb). REFpb uses the bank address supplied by the controller.

- REFab requires all banks idle and refreshes every bank; wait tRFCab before ACT or another refresh.
- REFpb refreshes the addressed bank; the bank must be idle. Wait tRFCpb before ACT or another REFpb to that bank; only tRRD is needed before ACT to a different bank. REFpb commands to different banks must satisfy tpbR2pbR.
- The controller tracks an internal bank counter synchronized by RESET_n, self-refresh exit, or any REFab. Repeating a REFpb to the same bank before all eight banks have been refreshed is illegal.
- Up to eight REFab commands may be postponed or pulled in (legacy mode); modified refresh limits apply at slower refresh rates. Per-bank refresh allows up to 8 x 8 per-bank commands to be postponed or pulled in, with a maximum of 2 x 8 x 8 within 2 x tREFI.

8.8 RFM - Refresh Management

RFM is required on some devices when the rolling accumulated ACT (RAA) counter reaches vendor thresholds.

- MR24 OP[0] = 1 indicates RFM is required. RAA increments per ACT per bank. When RAA reaches RAAIMT (MR24 OP[5:1]) additional refresh management is needed; it must never reach RAAMMT = RAAIMT x RAAMULT (MR24 OP[7:6]).
- RFMab decrements all banks' RAA by RAAIMT x RAADEC (MR36 OP[1:0]); RFMpb decrements only the selected bank by the same amount.
- RFM uses the REF encoding with CA3 = 1 when MR24 OP[0] = 1. RFM scheduling obeys the same minimum separations as REF.
- No RFM is required when the effective refresh interval tREFIe is at or below RFMTH = RAAIMT x tRC absolute min.

8.9 Self Refresh

- Entry: SRE with all banks idle and read data complete. CKE falls; only MRR-1, CAS-2, DES, SRX, MPC, MRW-1, and MRW-2 are allowed inside self refresh (except PASR and SR abort settings).
- Minimum stay is tSR = max(15 ns, 3 nCK). Exit to valid commands waits tXSR = max(tRFCab + 7.5 ns, 2 nCK).
- After any self-refresh exit, at least one extra refresh (one REFab or eight REFpb) must be issued before re-entering self refresh.
- Abort: if MR4 OP[3] is enabled, the DRAM aborts any ongoing refresh at SR exit and does not increment the refresh counter. The controller may issue a valid command after tXSR_abort = tRFCpb + 17.5 ns. The extra-refresh requirement still applies.

8.10 MRR / MRW / MPC

- MRR-1 + CAS-2 reads one of 64 mode registers. Valid OP code data appears on DQ[7:0] during the first 4 UI; DQS toggles for the full BL16 burst. Only DES is allowed during tMRR = 8 clocks.
- MRW is a two-cycle command (MRW-1 + MRW-2) carrying register address and data. tMRW = max(10 ns, 10 nCK); only DES is allowed during tMRW and tMRD = max(14 ns, 10 nCK). MRW is allowed from idle or active states.
- MPC provides NOP (OP6 = 0) and training functions. Write FIFO, Read FIFO, and Read DQ Calibration require CAS-2 immediately after MPC. Start/Stop DQS Oscillator and ZQCal Start/Latch do not require CAS-2 but need two additional DES/NOP cycles before the next command.

8.11 Power-down

- Entry: CKE falls. Illegal during RD, WR, MWR, MRR, MRW, VREF range/value MRW, command bus training MRW, VRCG high-current MRW, or Start DQS Oscillator. Allowed during ACT, PRE, auto-precharge, or REF.
- Idle power-down has no rows open; active power-down has at least one row open; self-refresh power-down continues internal refresh.
- Exit: CKE rises; first valid command after tXP = max(7.5 ns, 5 nCK). At least two clock transitions must occur during tXP.

8.12 Clock stop and frequency change

Clock stop or frequency change is allowed while CKE is low or while CKE is high with CS held low. When CKE is low, only REFab or REFpb may be executing and any ACT or PRE must have completed. When CKE is high, all commands and data bursts must have finished, RDA/WRA need four extra clocks, and normal operation resumes only after the clock is stable for at least 2 x tCK + tXP.

8.13 NOP / Deselect

DES (CS high) holds the bus idle for one cycle and never aborts an in-flight operation. MPC with OP6 = 0 is a two-cycle NOP.

Source: JESD209-4E sections 4.1, 4.2, 4.3, 4.4, 4.6, 4.10, 4.12, 4.13, 4.15, 4.16, 4.17, 4.18, 4.19, 4.20, 4.21, 4.22, 4.23, 4.24, 4.35, 4.44, 4.45, 4.47

9 Walkthrough: AXI Read, Row Already Open

Configuration used in all Chapter 4 traces (the drill model): RL = 10, WL = 6, BL = 16, flat 8-bank device, single channel view (channel A). Banks B0-B7, rows R0-R7, columns C0-C7. LPDDR4 is dual-channel per die; channel B runs on its own independent command/data bus and is not shown here.

9.1 Scenario

An AXI4 master issues a read that maps to bank B3, row R5, column C2. Row R5 is already open in B3, so the access is a row hit.

9.2 Command trace

```
RD  B3 C2         ; row already open - issue the burst immediately
NOP               ; tCCD = 8 clocks minimum between BL16 column
commands
NOP               ; read data returns after RL clocks plus tDQSCK
NOP               ; DQS may arrive over a window because there is no
DLL
NOP
NOP
NOP
NOP
NOP
```

The BL16 burst delivers 16 data words (D0-D15) across 8 clock cycles on the DQ bus. Because the drill model only has columns C0-C7, the burst wraps the 8-column address space twice.

9.3 Why each command

- `RD B3 C2` - the row is open, so no ACT is needed. The read is a two-cycle CA command; CA2r=H selects a read, C2 is the burst start column, and AP=0 leaves the row open.
- LPDDR4 supports BL16 and BL32; this trace uses BL16. A BL16 read transfers 16 contiguous words from the array. tCCD is 8 clocks for BL16.
- The first DQS edge appears RL clocks after the command, offset by tDQSCK. LPDDR4 has no DLL, so tDQSCK is a min-to-max window rather than a locked edge. The controller samples data relative to the strobe.
- No PRE: leaving the row open is the open-page policy; a later access to R5 in B3 is again a hit.

9.4 AXI view

- AR channel: one burst request. With a 16-bit DQ bus, BL16 covers 32 bytes in one DRAM burst.
- A wider AXI transaction would be split into multiple RD commands spaced tCCD apart, e.g. `RD B3 C2`, then `RD B3 C4` eight clocks later.

9.5 What to notice

- A row hit costs exactly two CA cycles on the command bus. Everything else (tCCD, turnaround) is about scheduling the next access around it.
- The controller tracks one open row per bank; that bookkeeping determines whether an AXI request becomes a hit or a miss.

Source: JESD209-4E sections 2.1, 4.1, 4.6, 4.7 (derived walkthrough)

10 Walkthrough: AXI Read, Wrong Row Open

Configuration used in all Chapter 4 traces (the drill model): RL = 10, WL = 6, BL = 16, flat 8-bank device, single channel view (channel A).

10.1 Scenario

An AXI4 read request maps to bank B3, row R6, column C0 - but row R5 is currently open in B3. This is a row conflict (miss): the wrong row must be closed before the requested row can be opened.

10.2 Command trace

```
PREpb B3          ; close the wrong row (tRAS from its ACT already met)
--- (tRPpb) ---   ; single-bank precharge time: bank unavailable
ACT B3 R6         ; open the requested row
--- (tRCD) ---    ; row-to-column delay
RD  B3 C0         ; burst read, AP=0, row stays open
NOP
NOP
NOP
NOP
NOP
NOP
NOP
```

10.3 Why each command

- `PREpb B3` - a bank holds exactly one open row. AB=0, so only B3 is precharged; every other bank is untouched. This is the per-bank precharge path, which uses the shorter tRPpb recovery.
- `--- (tRPpb) ---` - single-bank precharge time. A PREpb to one bank uses tRPpb; a PREab would cost tRPab and close banks that might still be doing useful work.
- `ACT B3 R6` - opens the requested row. The ACT-to-ACT spacing to B3 (prior ACT -> PREpb -> this ACT) satisfies tRC = tRAS + tRPpb by construction if tRAS was met before the PREpb.
- `--- (tRCD) ---` - row-to-column delay; the column circuitry cannot be driven until the row's data is in the sense amps.
- `RD B3 C0` - the read itself, identical to the hit case once the row is open. The BL16 burst wraps the C0-C7 drill columns twice.

10.4 AXI view

Same single AR request as the hit walkthrough - but the latency the master sees is now tRPpb + tRCD longer. Nothing on the AXI side changes; the penalty is entirely in DRAM scheduling.

10.5 What to notice

- Miss cost = tRPpb + tRCD before the first data can even start its RL journey.
- If the previous access to B3 had used AP=1, the bank would already be idle and only tRCD would be owed - that is the close-page trade in one line.
- PREpb may not be issued earlier than tRAS after the ACT that opened R5, nor earlier than tRTP after a read from R5. A scheduler that wants the earliest PREpb starts the tRAS/tRTP counters at the ACT/RD, not at the miss.

Source: JESD209-4E sections 4.1, 4.6, 4.17, 4.17.1 (derived walkthrough)

11 Walkthrough: AXI Write to an Idle Bank

Configuration used in all Chapter 4 traces (the drill model): RL = 10, WL = 6, BL = 16, flat 8-bank device, single channel view (channel A).

11.1 Scenario

An AXI4 write request maps to bank B1, row R2, column C0. B1 is idle (no open row).

11.2 Command trace

```
ACT B1 R2          ; open the row (bank was idle - no PRE needed)
--- (tRCD) ---     ; row-to-column delay
WR  B1 C0          ; burst write, AP=0; write data starts WL*tCK + tDQSS
later
NOP                ; burst occupies the DQ bus; tWR recovery runs from
NOP                ; the last data, then the bank may be precharged
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
```

The BL16 burst drives 16 data words (D0-D15) across 8 clocks, wrapping the C0-C7 drill model twice. LPDDR4 also supports masked writes via the Masked Write command and DMI pins; this trace uses a plain WR for simplicity.

11.3 Why each command

- `ACT B1 R2` - an idle bank needs no precharge; ACT is legal as long as tRRD from the previous ACT (any bank) and the tFAW window (max 4 ACTs in a rolling tFAW on 8-bank parts) are met.
- `--- (tRCD) ---` - same row-to-column delay as a read.
- `WR B1 C0` - the write is a two-cycle CA command. The controller drives DQS with a 2tCK preamble and the first data WL clocks (plus tDQSS) after the command. Each beat must meet tDS/tDH around the DQS edge. WDQS_on/off timing governs when DQS must be differential around the write; it can be ignored when it overlaps read bursts or turnarounds.
- The trailing NOPs cover tWR (write recovery from the end of the burst data): the bank may not be precharged until tWR has elapsed, because the write is still being committed to the array. The minimum WR-to-PRE spacing for the same bank is WL + 1 + BL/2 + RU(tWR/tCK) clocks.

11.4 AXI view

- AW and W channels carry the request; the DRAM does not see them separately. The controller issues WR only when it has (or is about to have) the write data, because LPDDR4 gives it no way to stall mid-burst.
- B (write response) can be returned once the data is accepted by the DRAM schedule; the tWR window is a DRAM-internal matter.

11.5 What to notice

- An idle-bank write miss costs tRCD, half the read-miss penalty - no tRP because nothing was open.
- Alternative: issue `WR B1 C0` with AP=1. The device then auto-precharges after the burst, and the bank returns to idle with no explicit PREpb - at the price of having programmed nWR correctly in MR1.
- tWR is the write-side analog of tRTP on the read side: both gate the PRE; both exist because the burst's last data and the array's precharge cannot overlap.

Source: JESD209-4E sections 4.1, 4.10, 4.13.2, 4.17.2 (derived walkthrough)

12 Walkthrough: Mixed Reads and Writes

Configuration used in all Chapter 4 traces (the drill model): RL = 10, WL = 6, BL = 16, flat 8-bank device, single channel view (channel A).

12.1 Scenario

Three AXI requests arrive close together:

1. Read: bank B3, row R5 (open), column C2.
2. Write: bank B1, row R2 (open), column C0.
3. Read: bank B1, row R2, column C4.

Both rows are already open, so this trace is pure bus scheduling: the DQ bus is shared, and read/write direction changes cost bubbles.

12.2 Command trace

```
RD  B3 C2         ; read burst 1: data out after RL + tDQSCK
NOP               ; read-to-write turnaround (book symbol tRTW)
NOP               ; formula: RL + RU(tDQSCKmax/tCK) + BL/2 - WL + tWPRE
+ RD(tRPST)
NOP               ; with drill values (2tCK preamble, 0.5nCK postamble,
tDQSCKmax=2tCK): 16 clocks
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
WR  B1 C0         ; write burst: DQ direction flips, data starts
WL+tDQSS later
NOP               ; write-to-read recovery (book symbol tWTR)
NOP               ; formula: WL + 1 + BL/2 + RU(tWTR/tCK)
NOP               ; with drill values: 6 + 1 + 8 + 6 = 21 clocks
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
NOP
RD  B1 C4         ; read burst 2, same open row as the write - a hit
NOP
```

```
NOP
NOP
NOP
NOP
NOP
NOP
PREpb B1         ; done with B1 (tRTP after the RD met); close it
```

12.3 Why each gap

- Read-to-write turnaround (book symbol tRTW): the write command may be issued no earlier than RL + RU(tDQSCK(MAX)/tCK) + BL/2 - WL + tWPRE + RD(tRPST) clocks after the read command. With RL=10, WL=6, BL=16, a 2tCK write preamble, 0.5nCK postamble, and a representative tDQSCK(MAX) of 2tCK, the gap is 16 clocks. The bubble exists because read data is still leaving the DRAM and the DQ bus must be idle before the controller starts driving write data. LPDDR4 has no DLL, so the controller must budget the worst-case tDQSCK(MAX).
- Write-to-read: the next read command waits WL + 1 + BL/2 + RU(tWTR/tCK) clocks after the write command (6 + 1 + 8 + 6 = 21 clocks here). tWTR lets the write data finish propagating into the array before sense amps are asked to read.
- `RD B1 C4` - same bank, same row as the write: a row hit even though a write happened in between. Open-row state is per bank, not per access type.
- `PREpb B1` - legal once tRAS (from the ACT that opened R2) and tRTP (from RD B1 C4) are both met. A PREab would have cost tRPab and closed every bank.

12.4 AXI view

AXI is oblivious to direction changes; its read and write channels are independent. The DRAM bus is not: every RD->WR or WR->RD transition on the shared DQ pins costs turnaround clocks, so a controller that groups reads with reads and writes with writes (batching) measurably outruns one that serves requests strictly in arrival order.

12.5 What to notice

- Turnarounds, not tRCD, dominate lightly-loaded mixed traffic.
- tCCD still applies between column commands: even without the direction change, bursts on the shared bus must not collide. With BL=16, tCCD is 8 clocks.
- tFAW limits activate parallelism: no more than four ACT or REFpb commands may fall inside a rolling tFAW window on this 8-bank device.
- Refresh appears as REFab in most schedules, but REFpb exists as a per-bank alternative that refreshes one bank while others remain active. LPDDR4 also adds Refresh Management (RFM): when the device advertises RFM support, the controller must issue RFM commands after a programmed number of ACTs to keep row-activation charge at safe levels.
- LPDDR4 is dual-channel per die. This trace follows channel A only; channel B runs independently on its own CA/DQ/DQS pins and can schedule its own reads and writes in parallel.
- Masked writes are available (Masked Write command + DMI pins), but this trace uses a plain WR for clarity.

Source: JESD209-4E sections 4.1, 4.2, 4.6, 4.10, 4.13.2, 4.17, 4.19, 4.47 (derived walkthrough)

13 Data Path

13.1 Prefetch and burst

LPDDR4 uses a 16n prefetch: each column access moves sixteen bits per DQ between the array and the IO, serialized as eight UI across two clock cycles. The default burst length is BL16. MR1 OP[1:0] can also select BL32 or BL16/32 on-the-fly, but there is no burst chop and no interleaved burst type. For reads, C[1:0] are implied zero, so the start column is always a multiple of four. For writes, C[3:2] must be driven low and the start address is aligned to the prefetch boundary (16n for BL16, 32n for BL32). In the drill model, BL16 delivers the eight columns C0-C7 twice.

13.2 Read and write strobes

One bidirectional differential pair (DQS_t/DQS_c) serves each byte lane. The DRAM drives it during reads and the controller drives it during writes.

Direction	Driver	Alignment	Preamble	Postamble
Read	DRAM	Edge-aligned to DQ	2 tCK (static or toggle)	0.5 tCK or 1.5 tCK
Write	Controller	Centered in DQ eye	2 tCK required	0.5 tCK or 1.5 tCK

Minimum preamble width is 1.8 tCK. The standard postamble is at least 0.4 tCK; the extended postamble is at least 1.4 tCK. There is no DLL. The read strobe access window is wide: tDQSCK spans 1.5 ns to 3.5 ns, with up to 4 ps/C and 7 ps/mV drift. The controller must train its capture logic to absorb this spread. DQS-to-DQ skew tDQSQ is at most 0.18 UI.

Symbol	Definition
tDQSCK	CK edge to first valid DQS edge: 1.5 ns min, 3.5 ns max
tDQSQ	DQS-to-DQ skew: max 0.18 UI
RD cmd -> first DQ	RL x tCK + tDQSCK + tDQSQ
WR cmd -> first DQS latching edge	WL x tCK + tDQSS
tDQSS	WR cmd CK edge to first DQS latching edge: 0.75-1.25 tCK
tDQS2DQ	DQS must arrive at DRAM before DQ by this trained offset
tDQSH / tDQSL	DQS high/low pulse width during writes: >= 0.4 tCK
tDSS / tDSH	DQS falling edge setup/hold to CK: >= 0.2 tCK

13.3 Read and write latencies

RL and WL are programmed through MR2. Two write-latency sets (A and B) trade extra latency for easier system timing at higher speeds; MR2 OP[6] selects the set. Enabling read DBI (MR3 OP[6]) adds two clocks to RL.

RL (no DBI / with DBI)	WL Set A	WL Set B	nWR	nRTP	Max CK freq
6 / 6	4	4	6	8	<= 266 MHz
10 / 12	6	8	10	8	<= 533 MHz
14 / 16	8	12	16	8	<= 800 MHz
20 / 22	10	18	20	8	<= 1066 MHz
24 / 28	12	22	24	10	<= 1333 MHz
28 / 32	14	26	30	12	<= 1600 MHz
32 / 36	16	30	34	14	<= 1866 MHz
36 / 40	18	34	40	16	<= 2133 MHz

The table above is for x16 mode; x8 byte mode uses slightly different codes.

Symbol	Turnaround relationship
tCCD	CAS-to-CAS delay: 8 tCK for BL16, 16 tCK for BL32
tWTR	WR -> RD: max(10 ns, 8 tCK), measured from the CK edge after the last write datum
tRTW	RD -> WR depends on DQ ODT state. With DQ ODT disabled: RL + RU(tDQSCK(max)/tCK) + BL/2 - WL + tWPRE + RD(tRPST). With DQ ODT enabled: RL + RU(tDQSCK(max)/tCK) + BL/2 + RD(tRPST) - ODTLon - RD(tODTon,min/tCK) + 1.

13.4 WDQS control

Before and after writes the differential DQS must keep enough voltage separation to keep the write receivers out of metastability. LPDDR4 offers two ways to manage this:

- Mode 1 (read-based): the SoC leaves DQS_c high except during read bursts, write bursts, and RD/WT turnarounds.
- Mode 2 (WDQS_on/off): after a Write-1 or Masked Write-1 command, DQS_t/DQS_c must be differential by WDQS_on (max) and can return to a don’t-care state after WDQS_off (min). These windows vary by WL set and frequency; overlap with read bursts or turnarounds is ignored.

Both modes cut DQS toggling to save power without changing normal command spacing.

13.5 Masked write, DM, and DBI

LPDDR4 has no dedicated DM pin. Each byte lane has a bidirectional DMI pin sampled with the DQ on both edges. What DMI carries depends on the operation and mode-register settings.

Function	MR bits	DMI meaning
Data mask for masked writes	MR13 OP[5] = 0, use Masked Write command	DMI high -> mask this beat; low -> write normally
Write DBI	MR3 OP[7] = 1	DMI high -> invert received DQ byte; low -> leave it
Read DBI	MR3 OP[6] = 1	DRAM drives DMI high when it inverts a read byte (more than four ones in the byte)

A Masked Write command is mandatory whenever any beat in the burst is masked; plain Write with DM disabled must drive DMI low. Masked Write supports only BL16; if the device is configured for BL32, the controller supplies only 16 bits of data for the masked write. Back-to-back masked writes to the same bank must be separated by tCCDMW = 32 tCK.

DBI limits the number of ones on the bus to cut power and improve DC balance. The DRAM decides read inversion per byte based on the bit count; the controller decides write inversion.

13.6 What LPDDR4 does not have

- No dedicated DM pin (the function moved to DMI).
- No parity on the CA bus.
- No write CRC; data integrity relies on DBI plus system-level schemes.
- No on-die ECC.
- No DLL, no burst chop, no interleaved burst order, no additive latency.

Source: JESD209-4E section 3, 3.4.1 (MR1, MR2, MR3, MR13), 4.4, 4.5, 4.6, 4.8, 4.9, 4.10, 4.12, 4.13, 4.14, 4.15, 4.16, Table 95, Table 100

14 On-Die Termination

LPDDR4 provides two independent termination domains. The CA-bus domain terminates the command/address signals and is controlled by a bond pad plus mode registers. The DQ domain terminates the data bus and is switched on and off by write commands. Both use a 240 ohm RZQ reference calibrated through the ZQ pin.

14.1 CA-bus ODT

CA ODT applies to CK_t, CK_c, CS, and CA[5:0]. It is enabled and its value is selected by MR11 OP[6:4]; the default is off. MR22 provides override bits: ODTE-CK (OP[3]) and ODTE-CS (OP[4]) can force termination on the clock or chip-select even when the ODT_CA bond pad is low, while ODTD-CA (OP[5]) can disable CA termination regardless of the pad.

ODT_CA pad	MR11 OP[6:4]	ODTD-CA	CA term	CK term	CS term
Low	Valid (not 000B)	0	Off	Off unless ODTE-CK=1	Off unless ODTE-CS=1
High	Valid	0	On	On	On
Any	Disabled or RFU	1	Off	On unless separately disabled	On unless separately disabled

A multi-rank system normally trains the terminating rank first, then the non-terminating rank. Unlike DQ ODT, CA ODT stays active through power-down and self-refresh. After writing a new CA ODT value, allow tODTUP before relying on the new setting; the spec leaves this update time as TBD.

14.2 DQ/DQS/DMI ODT

DQ ODT is selected by MR11 OP[2:0]; 000B disables it. Valid codes select RZQ/1 through RZQ/6. The DRAM leaves termination at high-Z until a Write-1 or Masked Write-1 command is sampled. It then turns on after a programmed ODTLon latency and turns off after ODTLoff once the write burst ends.

WL Set A	ODTLon (Set A)	ODTLoff (BL16, Set A)	Clock freq range
N/A	N/A	N/A	<= 266 / <= 533 MHz
8	4	20	533-800 MHz
12	4	22	800-1066 MHz
14	6	24	1066-1333 MHz
16	6	26	1333-1600 MHz
18	8	28	1600-1866 MHz
20	8	30	1866-2133 MHz

Set B uses larger ODTLon/off values; add 8 tCK to ODTLoff for BL32. The analog turn-on and turn-off windows are each 1.5 ns min to 3.5 ns max.

14.3 ODT during special operations

Situation	DQ/DQS/DMI ODT	CA ODT
Reads	Off / high-Z	As programmed
Write / Masked Write	On around burst per ODTLon/off	As programmed
Power-down / self-refresh	Feature off	Stays on if enabled
CA bus training	Follows MR11/MR22/ODT_CA pad	Follows MR11/MR22/ODT_CA pad
Write leveling	DQS on if enabled, DQ off	As programmed

During write leveling, DQS_t/DQS_c termination is on whenever DQ ODT is enabled, but the DQ lines remain unterminated.

14.4 ZQ calibration

The 240 ohm RZQ resistor calibrates both output driver strength and termination. Calibration is started with an MPC ZQCal Start command and the result is captured with an MPC ZQCal Latch command after tZQCAL (min 1 us). During tZQLAT (max(30 ns, 8 tCK)) the CA bus must be deselected so the new CA ODT value can settle. If the long calibration flow is not used, writing MR10 OP[0]=1 performs a ZQCal Reset that restores roughly +/- 30% accuracy.

14.5 What LPDDR4 does not have

- No dedicated ODT control pin for the DQ bus; termination turns on and off automatically around write commands.
- No VTT termination rail.
- No write CRC.
- No on-die ECC.

Source: JESD209-4E sections 3.4.1 (MR10, MR11, MR22), 4.38, 4.40, 4.41, 4.42, Table 159, Table 167, Table 168

15 Core Timing Parameters

The bank-state timings per channel (each LPDDR4 channel schedules independently). Values are from the core AC timing table; convert ns to clocks with RU(t/tCK).

15.1 Row-cycle parameters

Like LPDDR3, LPDDR4 carries separate per-bank and all-bank precharge times, and tRC is built from whichever one the history used:

Symbol	Definition	Applies between	Min
tRCD	Row-to-column delay	ACT(bank x) -> RD/WR(bank x)	max(18 ns, 4 nCK)
tRPpb	Precharge recovery, single bank	PRE(bank x) -> ACT(bank x)	max(18 ns, 4 nCK)
tRPab	Precharge recovery, all banks	PREA -> ACT(any)	max(21 ns, 4 nCK)
tRAS	Row-open window	ACT(bank x) -> PRE(bank x)	min max(42 ns, 3 nCK); max min(9 x tREFI x refresh rate, 70.2 us)
tRC	Full row cycle	ACT(bank x) -> ACT(bank x)	tRAS + tRPab, or tRAS + tRPpb

tPPD (precharge-to-precharge, 4 tCK) keeps back-to-back PRE commands off each other's heels; it does not apply to auto-precharges.

15.2 Activate-rate parameters

Symbol	Definition	Applies between	Min
tRRD	ACT to ACT, any two banks	ACT(bank x) -> ACT(bank y)	max(10 ns, 4 nCK)
tFAW	Four-activate rolling window	any 4 ACTs within a sliding window	40 ns (30 ns at 4267 Mb/s)

15.3 Data-to-precharge parameters

Symbol	Definition	Applies between	Min
tRTP	Internal analog read must finish before the row closes	last read data(bank x) -> PRE(bank x)	max(7.5 ns, 8 nCK)
tWR	Write recovery: last write data committed before precharge	last write data(bank x) -> PRE(bank x)	max(18 ns, 6 nCK) x16; max(20 ns, 6 nCK) x8

nRTP and nWR are the programmed clock equivalents (RU of the ns value); they set the auto-precharge start points after RD/WR with AP, and nRTP grows by 8 clocks for BL32 bursts.

Command-spacing formulas (same bank):

Transition	Earliest spacing (clocks)
RD -> PRE	BL/2 + max(8, RU(tRTP/tCK)) - 8
WR -> PRE	WL + 1 + BL/2 + RU(tWR/tCK)
RD/WRA with AP -> ACT	auto-precharge starts nRTP/nWR after the burst end; next ACT >= tRP (pb or ab) after it actually starts, and >= tRC

15.4 Why these numbers dominate

tRCD/tRP remain the row-miss penalty, with the per-bank/all-bank choice now routine: a controller that tracks open rows per bank will precharge one bank at a time and pay tRPpb almost always, keeping PREA for idle- drain events. tRRD/tFAW bound activate rate; the refresh-rate multiplier (MR4) shrinks both tRAS's maximum and the tREFI schedule together when the die runs hot or the vendor supports extended 1x windows.

Source: JESD209-4E sections 4.3 (Tables 88-89), 4.12, 4.17-4.19

16 Turnaround and Burst Timing

Column-command spacing and bus-direction rules. LPDDR4 is the first book in this series whose spec NAMES the read-to-write parameter: tRTW appears in the MPC/WDQS timing material.

16.1 Latencies and CAS-to-CAS

Read and write latency are programmed (no DLL), chosen per clock frequency from the latency tables, with separate columns for read DBI (+2 clocks when read DBI is enabled above the lowest rates) and two write-latency sets (MR2 OP6):

Symbol	Definition	Value
RL	Read latency, command to first data	6-36 nCK by frequency (10 at 266-533 MHz, 36 at 1866-2133 MHz; x8 mode runs higher)
WL	Write latency, command to first DQS	set A 4-18 nCK, set B 4-34 nCK, by frequency
nWR / nRTP	Programmed write-recovery / read-to-precharge clocks for auto-precharge	from the same table (RU of tWR/tRTP)
tCCD	Column command to column command	8 tCK (BL16); BL32 adds another 8

BL16 occupies 8 clocks of DQ - a much longer bus reservation than the DDR4-style 4. Seamless same-direction traffic issues a column command every 8 clocks, which is exactly tCCD. Burst interrupts are not allowed; BL32 may be selected on the fly for either direction.

16.2 Direction changes

Symbol	Definition	Value
tRTW	Read-to-write turnaround (NAMED in JESD209-4E): the loose read strobe (no DLL) plus the burst must clear before the write preamble	command gap = RL + RU(tDQSCK(MAX)/tCK) + BL/2 - WL + tWPRE + RU(tRPST); tRTW also appears in the MPC/WDQS timing table
tWTR	Write-to-read: last write data into the array before a read	max(10 ns, 8 nCK) x16; max(12 ns, 8 nCK) x8

Command-spacing forms (drill numbers in the Chapter 4 configuration: RL = 10, WL = 6, BL16, tWTR = 6 clk, tDQSCK(MAX) = 2 clk, tWPRE = 2 tCK, tRPST = 0.5 tCK):

Transition	Spacing (clocks)	Drill value
RD -> WR	RL + RU(tDQSCK(MAX)/tCK) + BL/2 - WL + tWPRE + RU(tRPST)	10 + 2 + 8 - 6 + 2 + 1 = 16 (the 0.5 nCK postamble rounds up)
WR -> RD	WL + 1 + BL/2 + RU(tWTR/tCK)	6 + 1 + 8 + 6 = 21
RD -> RD / WR -> WR	max(tCCD, BL/2) = 8	8

The no-DLL tDQSCK term is why tRTW is named and tabled in this spec: the controller must budget the worst-case strobe wander, not a locked skew. WDQS control (see the data-path chapter) can gate DQS drive between bursts, but its on/off constraints are checked against tRTW when ODT is disabled - the turnaround rules still win.

16.3 Strobe-level notes

- Read preamble: 2 tCK, static (no-toggle) or toggling per MR.
- Read postamble: 0.5 tCK standard, 1.5 tCK extended per MR.
- Write preamble 2 tCK; write postamble per the AC tables.
- tDQSCK: 1.5-3.5 ns min/max window (the value inside tRTW above); tDQSQ max 0.18 UI.
- DMI carries the mask (DM) and the DBI flag per byte lane; read DBI costs +2 RL clocks at the higher rates.

Source: JESD209-4E sections 4.5-4.13, 4.35 (MPC/WDQS timing), 10 (AC tables)

17 Refresh and Self-Refresh Timing

17.1 Refresh parameters

LPDDR4 keeps the LPDDR refresh-window model and doubles the command count: 8192 REFab per 32 ms window (versus LPDDR3's 4096), so the average interval is 3.904 us, not 7.8.

Symbol	Definition	Value
tREFW	Refresh window (1x rate)	32 ms; extensible at reduced refresh rates per MR4 OP[2:0]
R	Required REFab commands per window	8192 (all densities)
tREFI	Average REFab interval	3.904 us
tREFIpb	Average REFpb interval (per bank)	488 ns
tRFCab	All-bank refresh cycle	130 ns (1-2 Gb/ch), 180 (3-4), 280 (6-8), 380 (12-16)
tRFCpb	Per-bank refresh cycle	60 ns (1-2), 90 (3-4), 140 (6-8), 190 (12-16)
tpbR2pbR	REFpb(bank x) -> REFpb(bank y)	60 ns (1-4 Gb/ch), 90 (6 Gb/ch and up)

Rules:

- Each channel refreshes independently - a dual-channel die runs two independent refresh schedules (and the power delivery must tolerate both firing together).
- REFpb walks banks round-robin through an internal counter; the target bank is busy tRFCpb while the other seven keep working, and tRFCpb must be met before an ACT or another REFpb hits the same bank and before any REFab.
- The refresh rate (1x/0.5x/0.25x-style multipliers) is selected by MR4 OP[2:0]; higher-than-1x rates shrink tRAS's maximum and stretch tREFW. Above 85 C the rate must double (tREFI 1.95 us class).
- Refresh Management (RFM, sec 4.47) adds a second obligation: an internal rolling accumulated-ACT counter (per bank) must be decremented with

RFM commands before it saturates - activate-heavy traffic now has a refresh-like tax independent of tREFI.

17.2 Self-refresh

Symbol	Definition	Value
tSR	Minimum self-refresh time	max(15 ns, 3 nCK)
tXSR	Exit self-refresh to next valid command	max(tRFCab + 7.5 ns, 2 nCK)
tXSR_abort	Fast-exit abort path (high-density parts)	tRFCpb + 17.5 ns

Entry: all banks precharged, SRE encoding, clock may stop after the entry window. Exit: stable clock, CKE high, wait tXSR (or take the abort path on parts that offer it and re-issue the lost refresh). PASR (via MR) still restricts refresh to a fraction of the array at the cost of the rest.

17.3 Power-down

Symbol	Definition	Value
tXP	Exit power-down to any valid command	max(7.5 ns, 5 nCK)

Active and precharge power-down both exist; the array must still meet its refresh schedule across the stay. Deep power-down is gone from the user-visible command set compared with LPDDR3 - the modern mobile power story is FSP clock/frequency switching plus power-down, not a data-losing DPD state.

Source: JESD209-4E sections 4.19, 4.20, 4.21, 4.44, 4.47

18 Package and Electrical Notes

A light pass: what matters when connecting an LPDDR4 device, skipping the full ballout grids (spec sec 2 has many package variants).

18.1 Packages and channels

- LPDDR4 is dual-channel per die: two independent x16 channels (x8 byte mode selectable), each with its own CA bus, CK, CKE, CS, ODT and refresh schedule. Package variants pair, drop, or stack channels: quad-channel PoP FBGA (272-ball), two-channel discrete (200/203/254- ball classes), single-channel ePoP/MCP (144-ball), and wide x64 discrete packages for non-mobile density.
- 8 banks per channel, flat; densities per channel 1 Gb to 16 Gb (die totals to 32 Gb dual channel).
- Pins beyond LPDDR3: RESET_n (dedicated - reset is a pin, not an MRW), ODT(CA) per channel (CA-bus termination is new), DMI per byte lane (mask + DBI flag), and ZQ.

18.2 Supplies and signaling

Rail	Nominal	Feeds
VDD1	1.8 V	core/array
VDD2	1.1 V	core logic
VDDQ	1.1 V (0.6 V on 4X parts)	I/O buffers

LVSTL-class low-voltage signaling, unterminated by design; the CA bus gets its first termination option via ODT(CA) (MR-controlled). DQ ODT is asynchronous per MR, and DBI plus masked write do the power and partial-write jobs without a termination rail.

18.3 Speed grades

Data rates 533 to 4267 Mb/s per channel (tCK 3.75 ns down to ~0.468 ns per channel pair of Table 88 columns); the programmed RL/WL pair must be re-selected when the clock moves - which is what the frequency set points (FSP) exist for: two pre-programmed MR sets the controller switches between instead of retraining from scratch.

18.4 Characterization

IDD tables and pin capacitances live in the spec's later sections; consult them for power delivery (dual-channel simultaneous refresh is the worst case). Nothing there changes scheduling.

Source: JESD209-4E sections 2, 6.1, 4.3 (Table 88), 4.29

19 Test Features

LPDDR4 has no DDR4-style MPR pages, no device-level boundary scan, no OCD engine, no write CRC, and no gear-down mode. Instead it exposes a rich set of calibration and observation features reached through MRW/MRR and the MPC command. The two channels of a dual-channel die are largely independent, although they share the ZQ pin and calibration circuit.

19.1 VRCG high-current VREF generator

Set MR13 OP3 to 1 to engage the high-current VREF generator. This shortens the settling time of VREF(CA) and VREF(DQ) during training or when switching frequency set points. Only Deselect commands are legal until tVRCG_ENABLE elapses (max 200 ns). Clear MR13 OP3 to leave high-current mode and wait tVRCG_DISABLE (max 100 ns) before normal traffic. Range and value changes can still be done without VRCG, but they take longer.

19.2 CA and DQ VREF training

The CA receiver reference is tied to VDD2 and programmed in MR12, while the DQ receiver reference is tied to VDDQ and programmed in MR14. Because the two buses use different supplies, terminations, and swing environments, their reference levels are trained independently. Each register has a range bit (OP6) and a 6-bit value (OP[5:0]). One step is about 0.4% of the relevant supply. Wait the appropriate VREF step time between MRW updates.

Change type	Time
Single step	max 100 ns
Two or more steps in the same range	max 200 ns
Full range sweep	max 250 ns
Without VRCG high-current mode	max 1 ms

19.3 Command bus training

Enter Command Bus Training by setting MR13 OP0 to 1; after tMRD, drive CKE low. While CKE is low the DRAM uses the alternate FSP register copy; when CKE goes high it returns to the copy that was active before entry. Trained values are not retained, so write them to the FSP-WR registers before returning to the trained frequency.

For x16 devices, DQ[6:0] carries the desired VREF(CA) value, DQ6 selects the range, DQS0 toggles to latch it, and DQ[13:8] echoes the latched CA values. For byte (x8) devices there are two modes: Mode 1 echoes CA on DQ[5:0] and the VREF value is written beforehand via MRW; Mode 2 reuses DQ[5:0] first as VREF inputs and then as CA echoes, controlled by DQ7. In multi-rank systems train the terminating rank first, then turn on its termination while training the non-terminating rank.

Key timings include tCAENT (250 ns), tVREFCA_LONG (250 ns), tVREFCA_SHORT (80 ns for x16, 100 ns for byte Mode 2), tADR (20 ns), tXCBT (200 ns or 250 ns), and tMRZ (1.5 ns).

19.4 Frequency Set Points

Two complete copies of MR1, MR2, MR3, MR11, MR12, MR14, and MR22 are selected by MR13 OP6 (FSP-WR, which copy is read or written) and MR13 OP7 (FSP-OP, which copy is active). This lets the controller set up a new frequency, voltage swing, and termination point in the background, then switch atomically. To change FSP-OP, set MR13 OP7 and engage VRCG high-current at the same time, then wait tFC before resuming normal commands. tFC is 200 ns for a single VREF step or same-range changes and 250 ns when the VREF range changes. The device boots from FSP-OP[0] with un-terminated, low-frequency defaults.

19.5 Write leveling

Set MR2 OP7 to 1 to enter write-leveling mode. The DRAM samples CK on each rising DQS edge and asynchronously returns the sampled value on every DQ bit of that byte after tWLO. The controller adjusts the DQS delay until it sees a 0-to-1 transition. Both byte strobes in a channel are leveled independently, and dual-channel parts train each channel separately. Write leveling should be completed before DQS-DQ write training.

Symbol	Meaning	Value
tWLDQSEN	DQS quiet after mode entry	min 20 tCK
tWLMRD	First DQS edge after mode entry	min 20 tCK
tWLO	Feedback output delay	0 to 20 ns
tMRD	Exit MRW to valid command	min max(14 ns, 10 tCK)
tWLS/tWLH	DQS setup/hold to CK	speed-grade dependent

19.6 RD DQ calibration and read preamble training

Issue MPC-1 [RD DQ Calibration] followed immediately by CAS-2. The DRAM drives the MR32 pattern then the MR40 pattern on every DQ and DMI pin, using normal read-latency timing. MR15 and MR20 provide per-byte invert masks. Default values are MR32=5Ah, MR40=3Ch, and both masks=55h.

Read preamble training, enabled by MR13 OP1=1, holds DQS_t low and DQS_c high until the next RD DQ Calibration. This quiet strobe is useful for training the DQS receivers. Both features are optional; check the vendor data sheet.

19.7 DQS-DQ training

Because the internal DQS and DQ paths are not matched, the controller delays DQ relative to DQS to center the data eye. Training uses MPC [Write DQ FIFO] and [Read DQ FIFO], each followed immediately by CAS-2 and using normal write/read latencies. Up to five BL16 bursts can be written into the FIFO, then read back and compared against expected data.

The FIFO pointers reset at power-up, RESET_n assertion, power-down entry, or self-refresh power-down entry. To keep read and write pointers aligned, the number of Read FIFO commands must equal the number of Write FIFO commands modulo the FIFO depth of five.

19.8 DQS interval oscillator

A built-in ring oscillator copies the DQS clock tree to track delay drift caused by voltage and temperature changes. Start it with MPC [Start DQS Osc] and stop it with MPC [Stop DQS Osc] or an MR23 automatic-stop count; the result is stored in MR18 (LSB) and MR19 (MSB). Longer run times reduce granularity error, which is roughly 2 * tDQS2DQ / runtime. The count tells the controller whether retraining is needed and how large the timing error might be.

19.9 Multi-purpose command

MPC is a single opcode vehicle for training and calibration. OP6=0 is a NOP; OP6=1 selects a training function. Read/write functions (Write FIFO, Read FIFO, RD DQ Calibration) must be followed immediately by CAS-2 with operands driven low. Other MPC functions are Start/Stop DQS Oscillator, ZQCal Start, and ZQCal Latch.

19.10 Thermal offset and temperature sensor

MR4[6:5] applies a controller-supplied thermal offset to the channel's temperature-compensated self-refresh circuit; it can take up to 200 us to appear in MR4[2:0]. The on-die temperature sensor reports status through MR4 and updates at most every tTSI=32 ms. To avoid missing thermal events, the host polling interval must satisfy TempGradient * (ReadInterval + tTSI + SysRespDelay) <= 2C.

19.11 ZQ calibration

ZQ calibration is started with MPC [ZQCal Start] and captured with MPC [ZQCal Latch]. Allow tZQCAL=1 us before latching and keep CA deselected during tZQLAT. A ZQCal Reset via MR10 OP0 returns drivers to +/-30% accuracy in tZQRESET. Dual-channel devices share one ZQ pin and calibration circuit, but each channel latches its own values. The ZQ ball needs a 240 ohm +/-1% resistor to VDDQ and no more than 25 pF of loading.

19.12 Refresh management

Devices that need extra refresh under heavy traffic set MR24 OP0=1. The controller counts ACT commands per bank (RAA). When RAA reaches the vendor threshold RAAIMT in MR24[5:1], issue RFMab or RFMpb to give the DRAM extra refresh-management time. RFM decrements RAA by RAAIMT * RAADEC from MR36[1:0]; REF commands also decrement by RAAIMT. RAA must never exceed RAAMMT = RAAIMT * RAAMULT from MR24[7:6]; if it does, no further ACTs to that bank are legal until REF or RFM lowers it. Representative tRFMab/tRFMpb values are 210/170 ns for 8 Gb and 280/190 ns for 12/16 Gb.

19.13 Post-package repair

PPR is optional and repairs one fail row per bank per die by blowing an electrical fuse; status is readable in MR25. Procedure: precharge all banks; set MR4 OP4=1 and wait tMRD; ACT to the failing row; wait tPGM=1 s; PRE; wait tPGM_Exit=15 ns; clear MR4 OP4; wait tPGMPST=50 us; issue RESET and reinitialize. Only one row per bank can be repaired and the fuse cannot be undone. Memory contents are not refreshed during PPR and may be lost.

19.14 What is not here

- No MPR page concept; read training uses MRR and MPC-based RD DQ Calibration.
- No device-level boundary scan, no OCD calibration engine, no write CRC, no gear-down mode, and no per-pin loopback.

Source: JESD209-4E sections 4.25-4.38, 4.47, 4.48